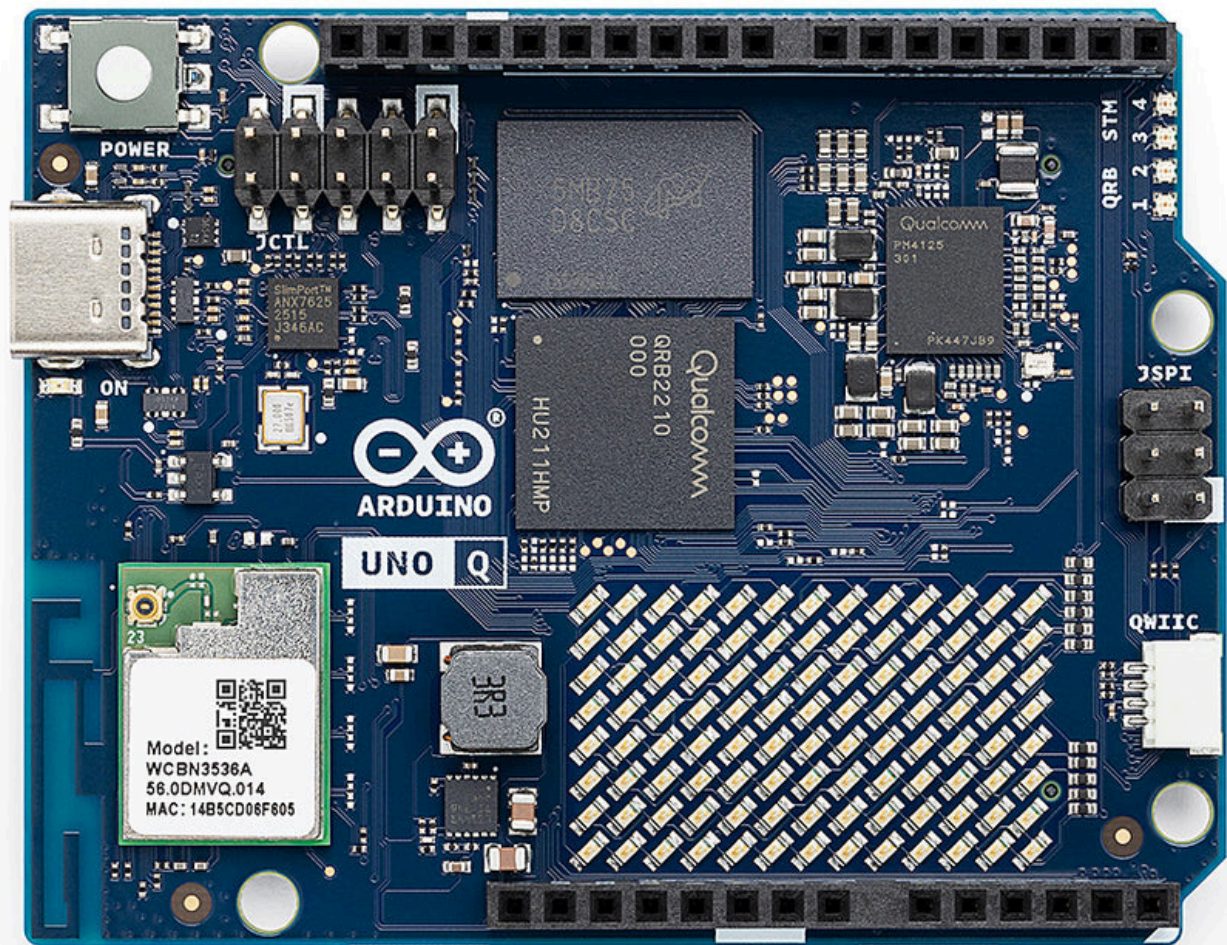


UNO Q Arduino Board

UNO Q board image:



Overview:

UNO Q is a dual-architecture Arduino board designed to bring higher performance, Linux computing, real-time control, wireless connectivity, and AI-ready development into the familiar UNO form factor. The board combines a Qualcomm Dragonwing QRB2210 microprocessor running a Debian-based Linux environment with an STM32U585 microcontroller built on the Arm Cortex-M33 architecture.

This hybrid design supports advanced embedded applications where high-level processing and deterministic hardware control need to work together. The Linux side handles application processing, Python workflows, AI models, networking, and multimedia tasks, while the microcontroller side manages Arduino sketches, sensors, actuators, and timing-sensitive control.

Positioning Statement:

UNO Q is a compact Arduino development board for edge AI, smart devices, robotics, connected products, and embedded Linux applications. It combines a high-performance application processor, a real-time microcontroller, Wi-Fi, Bluetooth, expansion connectors, and multi-environment development support in one board.

Core Architecture:

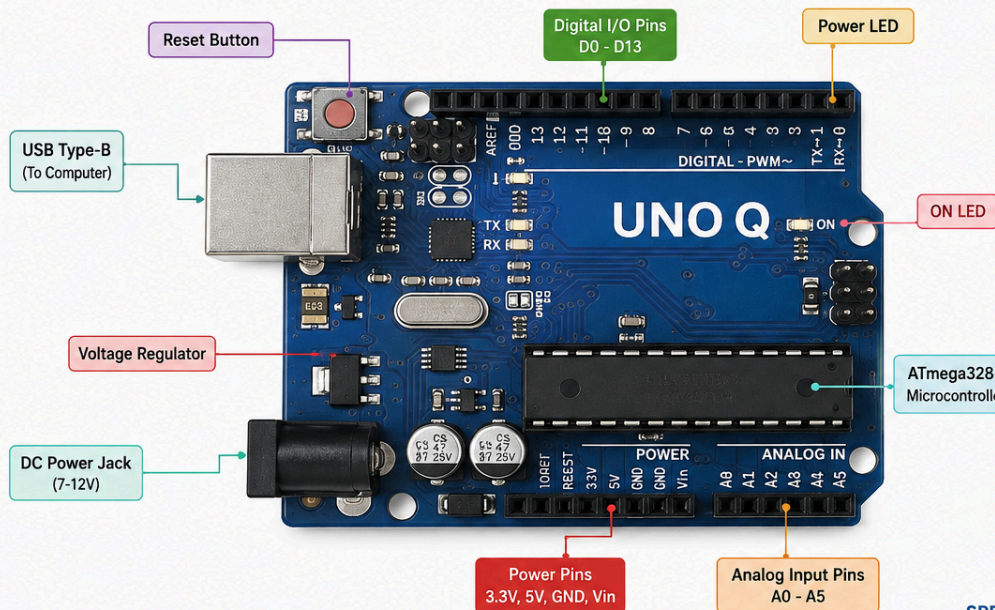
System	Component	Primary Function
Application processor	Qualcomm Dragonwing QRB2210	Runs Debian Linux, AI workloads, Python applications, networking, multimedia, and high-level software
Microcontroller	STM32U585 Arm Cortex-M33	Runs Arduino sketches over Zephyr OS and manages real-time control, sensors, actuators, and hardware timing
Wireless module	CYW55573	Provides Wi-Fi 5 and Bluetooth 5.1 connectivity
Expansion system	UNO headers, Grove connector, high-speed connectors	Supports shields, Modulino nodes, camera/display carriers, microSD, and custom hardware

Key Features:

- Dual-architecture design combining Linux computing and real-time microcontroller control
- Qualcomm Dragonwing QRB2210 microprocessor for high-performance processing
- Full Debian-based Linux environment with upstream support
- STM32U585 Arm Cortex-M33 microcontroller for Arduino sketches
- Arduino sketches supported over Zephyr OS
- Arduino App Lab support for Python apps, Arduino sketches, and AI models
- Arduino IDE 2.0 support for MCU development
- Built-in RPC library for communication between the MPU and MCU
- Wi-Fi 5 dual-band wireless connectivity
- Bluetooth 5.1 wireless connectivity
- Traditional UNO headers for shield compatibility
- Grove connector for Modulino node expansion
- High-speed connectors for advanced carriers and peripherals
- Single-board computer mode with compatible USB-C accessories
- Downloadable resources including pinout, schematics, STEP files, datasheet, and CAD files

Visual Feature:

UNO Q BOARD OVERVIEW



DOWNLOADABLE RESOURCES



Datasheet
Detailed technical specifications

[Download PDF](#)



User Manual
Getting started and usage guide

[Download PDF](#)



Example Code
Sample sketches and programs

[Download ZIP](#)



Schematic
Board schematic diagram

[Download PDF](#)



PCB Layout
Board PCB layout files

[Download ZIP](#)

KEY FEATURES



ATmega328P
Microcontroller



USB Programming
& Power



7-12V DC
Input Voltage



14 Digital I/O Pins
(6 PWM)



6 Analog
Input Pins



On-board
LED Indicators

UNO Q – Perfect for learning, prototyping and building amazing projects!

SPECIFICATIONS

- Microcontroller : ATmega328P
- Operating Voltage : 5V
- Input Voltage (recommended) : 7-12V
- Digital I/O Pins : 14 (of which 6 provide PWM output)
- Analog Input Pins : 6
- Flash Memory : 32 KB (ATmega328P)
- SRAM : 2 KB (ATmega328P)
- EEPROM : 1 KB (ATmega328P)
- Clock Speed : 16 MHz

Featured Area	Details Shown
Product identity	UNO Q board name and product image
Main description	Dual-architecture Arduino board for performance, Linux, and real-time control
Processor combination	Qualcomm Dragonwing QRB2210 MPU and STM32U585 MCU
Operating environments	Debian Linux and Arduino sketches over Zephyr OS
Resources	Pinout PDF, schematics, STEP files, datasheet, CAD files
Feature tabs	Features, tutorials, tech specs, compatibility, suggested libraries

Technical Specifications:

Category	Specification
Board	UNO Q
Form factor	UNO-style compact board
Main processor	Qualcomm Dragonwing QRB2210
CPU	Quad-core Arm Cortex-A53 up to 2.0GHz
GPU	Adreno GPU
Application OS	Debian Linux
Microcontroller	STM32U585
MCU architecture	Arm Cortex-M33

MCU clock speed	Up to 160MHz
MCU flash memory	2MB
MCU SRAM	786KB
MCU math support	Floating Point Unit
RAM shown	2GB RAM
Storage shown	16GB eMMC
Wireless module	CYW55573
Wi-Fi	Wi-Fi 5 dual-band 2.4GHz/5GHz
Bluetooth	Bluetooth 5.1
Development tools	Arduino App Lab, Arduino IDE 2.0, Arduino CLI-compatible workflows
Interprocessor communication	Built-in RPC library
Expansion	UNO headers, Grove connector, high-speed board connectors

Qualcomm Dragonwing QRB2210 Microprocessor

The Qualcomm Dragonwing QRB2210 microprocessor provides the high-performance computing side of UNO Q. It enables Linux-based applications, AI workloads, graphics acceleration, networking, and complex data processing.

Specification	Detail
Processor	Qualcomm Dragonwing QRB2210
CPU architecture	Quad-core Arm Cortex-A53
Maximum CPU speed	2.0GHz
Graphics	Adreno GPU
Memory shown	2GB RAM
Storage shown	16GB eMMC
Math capability	Floating Point Unit
Operating environment	Debian Linux

MPU Capabilities:

- Linux application development
- Python application execution
- Edge AI model workflows
- Computer vision processing
- Sound and audio intelligence processing
- Wireless networking services
- Data processing and storage
- Multimedia handling
- Communication with the MCU through RPC
- Single-board computer-style workflows

STM32U585 Arm Cortex-M33 Microcontroller

The STM32U585 microcontroller provides the real-time control side of UNO Q. It is designed for deterministic embedded tasks such as reading sensors, controlling actuators, driving outputs, and running Arduino sketches.

Specification	Detail
Microcontroller	STM32U585
Core	Arm Cortex-M33
Maximum clock speed	160MHz
Flash memory	2MB
SRAM	786KB
Math capability	Floating Point Unit
Sketch runtime	Arduino sketches over Zephyr OS

MCU Capabilities:

- Arduino sketch execution
- Sensor polling and signal reading
- Actuator control
- Motor control logic
- LED and display control
- Time-sensitive hardware operations
- Low-level peripheral management
- Deterministic embedded responses

Dual-Architecture Advantage:

UNO Q separates workloads across two processing environments. The Linux application processor handles complex computation, while the microcontroller handles real-time physical control.

Workload Type	Recommended Processing Side	Reason
AI inference	MPU	Requires higher processing capability and Linux-based software support
Computer vision	MPU	Benefits from Linux tools, camera support, and accelerated processing
Python application logic	MPU	Runs naturally in the Debian-based environment
Sensor timing	MCU	Requires predictable real-time behavior
Motor control	MCU	Requires fast, deterministic hardware response
Network communication	MPU	Benefits from Linux networking support
Physical actuation	MCU	Requires reliable output timing
Hybrid automation	MPU and MCU	Combines intelligent decision-making with real-time control

Debian Linux Environment:

UNO Q includes a Debian Linux operating system powered by the application processor. This gives

the board a native Linux development environment for building advanced embedded applications.

Debian Linux Benefits:

- Full Linux software environment
- Support for high-level application development
- Python-based workflows
- AI and machine learning experimentation
- Networking and device communication
- Local data processing
- Single-board computer usage
- Application development without relying only on microcontroller workflows

Arduino App Lab Support:

Arduino App Lab supports development across the dual-architecture system. It allows Python applications, Arduino sketches, and AI models to work together in a unified workflow.

Arduino App Lab Enables:

- Python application development
- Arduino sketch development
- AI model integration
- MPU and MCU coordination
- Hybrid embedded workflows
- Edge AI experimentation
- Physical computing with intelligent processing

Arduino IDE 2.0 Compatibility:

UNO Q supports Arduino IDE 2.0 for MCU-side development. This preserves the familiar Arduino workflow while extending the board into more advanced Linux and AI applications.

Arduino IDE 2.0 Uses:

- Programming the STM32U585 MCU
- Running Arduino sketches over Zephyr OS
- Building sensor and actuator projects
- Prototyping traditional Arduino applications
- Working with embedded hardware interfaces

Arduino CLI and Editor Workflows:

UNO Q supports development approaches compatible with Arduino CLI and popular editor-based workflows. This allows flexible development for embedded projects, automation scripts, and advanced software pipelines.

Workflow Options:

Workflow	Best Use
Arduino App Lab	Hybrid projects combining Python, AI models, and Arduino sketches
Arduino IDE 2.0	MCU programming and traditional Arduino sketch development

Arduino CLI-compatible workflows	Scripted builds, terminal-based development, and custom workflows
Editor-based workflows	Advanced software development and project organization

Built-In Remote Procedure Call:

UNO Q includes a built-in RPC library that bridges the Linux MPU and the MCU. This allows the two processing systems to communicate and coordinate tasks.

RPC Use Examples:

Linux MPU Task	MCU Task	Example Application
Process image input	Trigger an actuator	AI camera lock system
Analyze sound data	Control LED effects	Sound-reactive lighting
Receive wireless command	Move a motor	Connected robot
Run AI model	Activate alert output	Edge AI safety monitor
Log sensor data	Read sensor values	Smart environmental station
Host local interface	Control hardware state	Interactive control panel

Wireless Connectivity

UNO Q includes the CYW55573 wireless module for Wi-Fi and Bluetooth communication.

Wireless Feature	Specification
Wireless module	CYW55573
Wi-Fi standard	Wi-Fi 5
Wi-Fi bands	2.4GHz and 5GHz dual-band
Wi-Fi antenna	Onboard antenna
Bluetooth version	Bluetooth 5.1
Bluetooth antenna	Onboard antenna

Wireless Project Applications

- Connected IoT devices
- Remote monitoring systems
- Smart home prototypes
- Wireless sensor networks
- Bluetooth-connected peripherals
- Cloud-connected edge devices
- Networked robotics
- Local device dashboards

Single-Board Computer Mode:

UNO Q can operate as a single-board computer through its Debian Linux environment. With a USB-C dongle that supports external power delivery connected to the UNO Q USB-C port, the board can

connect to standard computing peripherals.

Compatible Peripheral Examples:

- HDMI display
- Mouse and keyboard
- USB camera
- USB drive3
- Ethernet cable
- microSD card
- USB microphone
- USB headphones

Recommended SBC Configuration:

The 4GB RAM UNO Q variant is recommended for single-board computer usage.

Expansion and Hardware Compatibility:

UNO Q includes multiple expansion options for modular hardware development and custom prototyping.

Expansion Interface	Purpose
Grove connector	Connects Modulino nodes and compatible modular components
Traditional UNO headers	Supports official and custom Arduino shields
Two bottom high-speed connectors	Support carriers for advanced peripherals
Carrier compatibility	Allows connection of MIPI CSI cameras, MIPI DSI displays, microSD cards, and additional hardware



DOWNLOADABLE RESOURCES

All the essential hardware resources for UNO Q development board.



1. DATASHEET

Detailed technical specifications of UNO Q board, electrical characteristics and mechanical dimensions.

 PDF File | 1.25 MB

Download PDF



2. USER MANUAL

Complete guide to get started with UNO Q. Includes board overview, pin description, setup instructions, and example projects.

 PDF File | 2.35 MB

Download PDF



3. EXAMPLE CODE

Sample Arduino sketch codes for different applications like LED, sensors, motors, Bluetooth, LCD, and more.

 ZIP File | 4.48 MB

Download ZIP



4. SCHEMATIC

Complete electronic circuit schematic diagram of the UNO Q board.

 PDF File | 1.10 MB

Download PDF



5. PCB LAYOUT

PCB design files including component placement, copper layers, and board outline.

 ZIP File | 8.75 MB

Download ZIP

Resource	Purpose
Pinout PDF	Identifies board pins, signals, and hardware connections
Schematics	Shows electrical design and circuit-level information
STEP files	Supports mechanical planning, product design, and enclosure work
Datasheet	Provides technical details, ratings, and specifications
CAD files	Supports hardware integration and design workflows

Feature Tabs Shown in Reference Screen

Tab	Likely Content Area
Features	Board capabilities, architecture, wireless connectivity, expansion, and development workflows
Tutorials	Guides for setup and project development
Tech Specs	Detailed hardware and electrical specifications
Compatibility	Supported software, hardware, and accessories
Suggested Libraries	Recommended libraries for development workflows

Capabilities by Application Area:

Edge AI

UNO Q supports edge AI projects by combining Linux-based processing with real-time hardware control. AI models can run on the MPU while the MCU manages physical inputs and outputs.

Computer Vision

The application processor supports image processing and vision workflows, while expansion options can support camera-related hardware through compatible carriers.

Sound Intelligence

USB microphone and audio peripheral support make UNO Q suitable for sound-reactive and audio-processing applications.

Robotics

The Linux side can manage navigation logic, perception, and decision-making, while the microcontroller handles motors, encoders, sensors, and timing-sensitive movement.

Smart Home

Wi-Fi, Bluetooth, Linux applications, and real-time control make UNO Q suitable for smart home automation, connected sensing, lighting control, and monitoring systems.

IoT Prototyping

The board supports wireless communication, local data processing, cloud-connected workflows, and sensor-driven automation.

Education

UNO Q supports a learning path from Arduino sketches to Linux computing, AI integration, and advanced embedded systems.

Professional Prototyping

The board provides a compact development platform for testing connected products, intelligent devices, automation systems, and interactive hardware.

Example Project Concepts

AI Vision Door Monitor

A camera captures image data, the Linux side processes visual information, and the MCU controls alerts, lights, relays, or locks.

Smart Environmental Station

Sensors collect environmental readings through the MCU, while the Linux side logs data, displays insights, and manages network communication.

Sound-Reactive Lighting Controller

A USB microphone captures audio, the MPU analyzes sound patterns, and the MCU drives LED effects with real-time responsiveness.

Autonomous Mini Robot

The MPU handles navigation logic or vision-based decisions, while the MCU controls motors, distance sensors, and motion timing.

Connected Industrial Sensor Node

The MCU collects sensor data from physical equipment, while the MPU performs local processing, wireless communication, and event reporting.

AI-Powered Object Sorter

The Linux side classifies objects using a camera or sensor input, while the microcontroller controls servos, motors, or gates for sorting actions.

Interactive Smart Display

The MPU runs a Linux-based interface, while the MCU manages buttons, sensors, LEDs, and real-time feedback.

System Workflow Example

Step	System Action	Processing Side
1	Sensor, camera, or microphone captures input	MCU or MPU
2	Data is processed locally	MPU
3	AI model, Python app, or Linux service makes a decision	MPU
4	Decision is sent through RPC	MPU to MCU
5	Hardware output is activated	MCU
6	Status is logged, displayed, or transmitted wirelessly	MPU

Certifications and Regulatory

The reference screenshots show the following certification and compliance marks associated with UNO Q:

Certification/Mark
WPC
Anatel
CASA
INMA
NCC
CE
UKCA
KC
FCC DSS
FCC DTS
FCC NII
ISED
SRRC
OFCA